

Ques 1

Section - A (Analog Design)
 ADVD Compore Exam 2014 I Sem.

Given:- To design a current sink of $200 \mu A$, & $V_{min} = 0$
 i.e. $I_{out} = 200 \mu A$, given $I_{ref} = 50 \mu A$

$$\left(\frac{W}{L}\right)_7 = 2/1, \text{ hence } \left(\frac{W}{L}\right)_6 = 8/1 \quad \text{--- (1m)}$$

$$\text{now } I_{in} = 200 \mu A.$$

$$\begin{aligned} V_{GS5} = V_{G2} &= V_t + V_{ov5} \\ &= 0.7 + \sqrt{\frac{2 \cdot 50}{140 \left(\frac{W}{L}\right)_5}} \quad \text{--- (i)} \end{aligned}$$

$$\begin{aligned} \text{from } V_{min} &= 0.6 V = 2V_{ov1} \\ V_{ov1} &= V_{ovL} = 0.3 V \end{aligned}$$

$$\begin{aligned} \text{from figure } V_{G2} &= V_{GS2} + V_{ov1} \\ V_{G2} &= V_t + 2V_{ov1} \quad \text{--- (ii)} \end{aligned}$$

equating (i) & (ii)

$$\begin{aligned} \sqrt{\frac{100}{140 \left(\frac{W}{L}\right)_5}} &= 0.6 \\ \left(\frac{W}{L}\right)_5 &= 1.984 \approx 2 \quad \rightarrow (3m) \end{aligned}$$

$$\text{Similarly } V_{ov1} = V_{ov2} = 0.3$$

$$\begin{aligned} \sqrt{\frac{2 \cdot 200}{140 \left(\frac{W}{L}\right)_{1-4}}} &= 0.3 \\ \left(\frac{W}{L}\right)_{1-4} &= \frac{40}{14 \times 0.09} = 31.75 \approx 32 \quad \rightarrow (3m) \end{aligned}$$

$$(i) \quad R_{out} = g_{m2} r_{o2} r_{o1}, \quad r_{o1} = r_{o2} = \frac{1}{0.1 \times 200 \times 10^{-6}} = 50 K$$

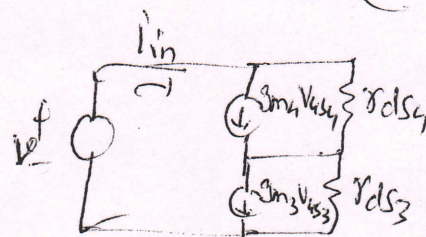
$$g_{m2} = \sqrt{2 \cdot 200 \times 10^{-6} \times 31.75 \times 140 \times 10^{-6}} = 1.33 \text{ mA/V} \quad \rightarrow (1m)$$

$$R_{out} = 1.33 \times 2500 K = 3.325 \text{ m}\Omega \quad \rightarrow (2m)$$

(ii) for the shaded region:-

$$V_{GS4} = -V_{S4}, \quad V_{GS3} = V_{G3} = 0$$

$$\begin{aligned} 0 &= r_{ds4} (I_{in} - g_{m4} V_{GS4}) + V_x \\ &= r_{ds4} (I_{in} + g_{m4} V_{GS4}) + V_x \end{aligned}$$



$$v = i_{in} r_{ds4} + (i_{in} - g_{m3} v) r_{ds3} (1 + g_{m4} r_{ds4})$$

$$\left. \begin{aligned} \lim_{r_{ds4} \rightarrow \infty} r_{ds4} = r_{ds3} \rightarrow \infty \end{aligned} \right\} \frac{v}{i_{in}} = \frac{r_{ds4} + r_{ds3} + g_{m4} r_{ds3} r_{ds4}}{1 + g_{m3} r_{ds3} (1 + g_{m4} r_{ds4})}$$

$$\text{So } \frac{v}{i_{in}} = \frac{g_{m4}}{g_{m3} g_{m4}} = \frac{1}{g_{m3}} \quad \text{--- (4m)}$$

(You can even ignore r_{ds} initially in figure to simplify needed equation)

(4) Advantage : "wide swing current sink" --- (1m)

Q3:

$$I_{SS} = I_{CM1} = \frac{5 \text{ mW}}{3.3} = 1.52 \text{ mA} \quad (1 \text{ M})$$

$$I_{d1}(M2) = I_{d1}(M3) = 0.76 \text{ mA}$$

$$\begin{aligned} V_{OV}(M4-M7) &= \sqrt{\frac{2 \times 0.76 \times 10^{-3}}{140 \times 10^{-6} \times 100}} \\ &= 0.329 \text{ V} \\ &= V_{OV}(M9) \end{aligned}$$

$$\begin{aligned} V_Z &= V_{GS7} = V_{tn} + V_{OV, n} \\ &= V_{out1} = 0.7 + 0.329 = 1.029 \text{ V} \quad (1 \text{ M}) \end{aligned}$$

(ii)

$$\begin{aligned} g_{m2} &= \frac{2I_D}{V_{OV}(M2)} = \frac{2 \times 0.76}{0.6} \\ &= 2.53 \text{ mA/V} \end{aligned}$$

$$r_{o2} = \frac{1}{\lambda P_{D2}} = \frac{1}{0.1 \times 0.76 \text{ m}} = 13.157 \text{ k}$$

$$\begin{aligned} A_1 &= \frac{V_{out1}}{V_{in}} = g_{m2} (r_{o2} \parallel g_{m4} r_{o4} r_{o6}) \\ &\approx g_{m2} r_{o2} \approx 33.5 \text{ V/V} \quad (1 \text{ M}) \end{aligned}$$

For CSA:

$$\begin{aligned} g_{m8} &= \sqrt{2 \times 40 \times 10^{-6} \times 100 \times 0.76 \times 10^{-3}} \\ &= 2.46 \text{ mA/V} \end{aligned}$$

$$r_{o8} = r_{o9} = 13.157 \text{ k}$$

$$\begin{aligned} A_2 &= \frac{V_{out2}}{V_{out1}} = -g_{m8} (r_{o8} \parallel r_{o9}) \\ &= -16.18 \text{ V/V} \quad (1 \text{ M}) \end{aligned}$$

$$A = A_1 \times A_2 = -542.13 \text{ V/V} \quad (1 \text{ M})$$

(iii)

$$|W_P(v_{out1})| \approx \frac{1}{R_{out1} \times C_{eq}(v_{out1})}$$

$$R_{out1} = 13.157 \text{ k}$$

$$C_{eq}(v_{out1}) = 1918 \text{ fF}$$

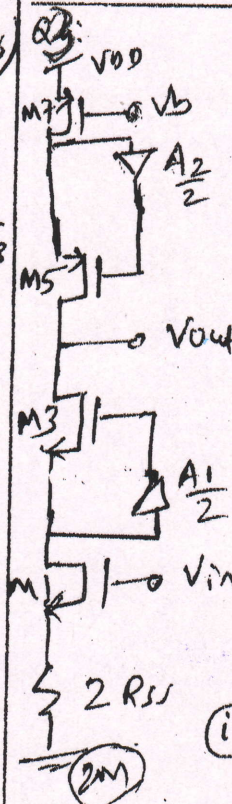
$$|W_P(v_{out1})| \approx \frac{39.62 \text{ M}}{(6.30 \text{ MHz})} \quad (2 \text{ M})$$

$$|W_P(v_{out2})| \approx \frac{1}{R_{out2} (C_{Rat2})}$$

$$\begin{aligned} C_{Rat2} &= C_L + C_c + (C_{db8} + C_{db9}) \\ &= [500 + 100 + 50] \text{ fF} \\ &= 650 \text{ fF} \end{aligned}$$

$$\begin{aligned} |W_P(v_{out2})| &\approx \frac{1}{6.578 \text{ k} \times 650 \times 10^{-15}} \\ &= 233.87 \text{ M} \quad (2 \text{ M}) \\ &\quad (37.22 \text{ MHz}) \end{aligned}$$

$$\begin{aligned} |W_2| &\approx \frac{g_{m8}}{C_c} \\ &= \frac{2.46 \times 10^{-3}}{100 \times 10^{-15}} \quad (1 \text{ M}) \\ &= 24600 \text{ Mrad/sec} \\ &\quad (3915.21 \text{ MHz}) \end{aligned}$$



(i)

$$R_{o1} \approx (A_2 g_{m5} r_{o5} r_{o7} \parallel A_1 g_{m3} r_{o3} r_{o1})$$

$$\frac{V_{out1}}{V_{in}} \approx -g_{m1} R_{o1} \quad (2 \text{ M})$$

$$R_{o2} \approx (A_2 g_{m6} r_{o6} r_{o8} \parallel A_1 g_{m4} r_{o4} r_{o2})$$

$$\frac{V_{out2}}{V_{in}} \approx g_{m2} R_{o2} \quad (2 \text{ M})$$

$$\frac{V_{out2} - V_{out1}}{V_{in}} = g_{m2} R_{o2} - g_{m1} R_{o1} \quad (1 \text{ M})$$

$$(ii) A_{cm} \approx \frac{g_{m1}}{1 + g_{m1} (2R_{SS})} \quad (1 \text{ M})$$

$$R_{up} \approx A_2 \frac{1}{2} g_{m5} r_{o5} r_{o7} \quad (1 \text{ M})$$

$$R_{down} \approx A_1 \frac{1}{2} g_{m3} g_{m1} r_{o3} r_{o1} \quad (1 \text{ M})$$

Q3:

$$(i) I_{SS} = I_{CM1} = \frac{5 \text{ mW}}{3.3} = 1.52 \text{ mA}$$

$$I_{d1}(M2) = I_{d1}(M3) = 0.76 \text{ mA} \quad (1 \text{ M})$$

$$\begin{aligned} V_{OV}(M4-M7) &= \sqrt{\frac{2 \times 0.76 \times 10^{-3}}{140 \times 10^{-6} \times 100}} \\ &= 0.329 \text{ V} \\ &= V_{OVCM9} \end{aligned}$$

$$\begin{aligned} V_Z &= V_{GS7} = V_{tn} + V_{OV, n} \\ &= V_{out1} = 0.7 + 0.329 = 1.029 \text{ V} \end{aligned} \quad (1 \text{ M})$$

(ii)

$$\begin{aligned} g_{m2} &= \frac{2I_{D2}}{V_{OVCM2}} = \frac{2 \times 0.76}{0.6} \\ &= 2.53 \text{ mA/V} \end{aligned}$$

$$r_{O2} = \frac{1}{\lambda P_{D2}} = \frac{1}{0.1 \times 0.76 \text{ m}} = 13.157 \text{ K}$$

$$A_1 = \frac{V_{out1}}{V_{in}} = g_{m2}(r_{O2} \parallel g_{m4} r_{O4} r_{O6})$$

$$\approx g_{m2} r_{O2} \approx 33.5 \text{ V/V} \quad (1 \text{ M})$$

For CSA:

$$\begin{aligned} g_{m8} &= \sqrt{2 \times 40 \times 10^{-6} \times 100 \times 0.76 \times 10^{-3}} \\ &= 2.46 \text{ mA/V} \end{aligned}$$

$$r_{O8} = r_{O9} = 13.157 \text{ K}$$

$$\begin{aligned} A_2 &= \frac{V_{out2}}{V_{out1}} = -g_{m8}(r_{O8} \parallel r_{O9}) \\ &= -16.18 \text{ V/V} \quad (1 \text{ M}) \end{aligned}$$

$$A = A_1 \times A_2 = -542.13 \text{ V/V} \quad (1 \text{ M})$$

(iii)

$$|W_P(v_{out1})| \approx \frac{1}{R_{out1} \times C_{eq}(v_{out1})}$$

$$R_{out1} = 13.157 \text{ K}$$

$$C_{eq}(v_{out1}) = 1918 \text{ fF}$$

$$|W_P(v_{out1})| \approx 39.62 \text{ M}$$

$$|W_P(v_{out2})| \approx \frac{1}{R_{out2} \times C_{eq}(v_{out2})} \quad (6.30 \text{ MHz})$$

$$R_{out2} = C_{out2}$$

$$\begin{aligned} C_{out2} &= C_L + C_C + (C_{db8} + C_{db7}) \\ &= [500 + 100 + 50] \text{ fF} \\ &= 650 \text{ fF} \end{aligned}$$

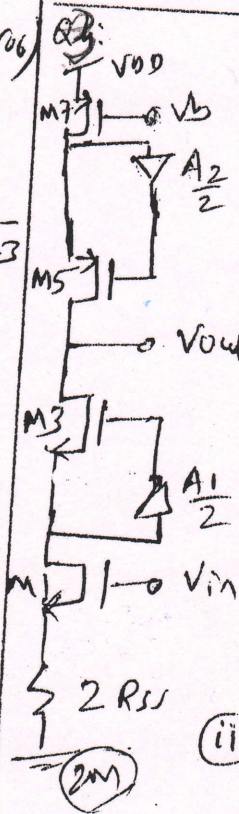
$$\begin{aligned} |W_P(v_{out2})| &\approx \frac{1}{6.578 \text{ K} \times 650 \times 10^{-15}} \\ &= 233.87 \text{ M} \quad (37.22 \text{ MHz}) \end{aligned}$$

$$|W_2| \approx \frac{g_{m8}}{C_C}$$

$$= \frac{2.46 \times 10^{-3}}{100 \times 10^{-15}} \quad (1)$$

$$= 24600 \text{ Mrad/sec}$$

$$(3915.21 \text{ MHz})$$



(i)

$$R_{O1} \approx (A_2 g_{m5} r_{O5} r_{O7} \parallel A_1 g_{m3} r_{O3} r_{O1})$$

$$\frac{V_{out1}}{V_{in}} \approx -g_{m1} R_{O1}$$

$$R_{O2} \approx (A_2 g_{m6} r_{O6} r_{O8} \parallel A_1 g_{m4} r_{O4} r_{O2})$$

$$\frac{V_{out2}}{V_{in}} \approx g_{m2} R_{O2} \quad (2 \text{ M})$$

$$\frac{V_{out2} - V_{out1}}{V_{in}} = g_{m2} R_{O2} - g_{m1} R_{O1} \quad (1 \text{ M})$$

$$(ii) A_{CM} \approx \frac{g_{m1}}{1 + g_{m1}(2R_{SS})} \quad (1 \text{ M})$$

$$R_{up} \approx A_2 \frac{1}{2} g_{m5} r_{O5} r_{O7} \quad (1 \text{ M})$$

$$R_{down} \approx A_1 \frac{1}{2} g_{m3} g_{m1} r_{O3} r_{O1} \quad (1 \text{ M})$$

Q.4.

(a) Min. clock cycle time.

We need to check for all paths (with skew/without skew)

$$T_{\text{cycle min}} \geq T_{c-q} + T_{\text{logic}} + T_{s.u} \quad \text{2.5}$$

$$t_{\text{hold}} < t_{c-q} + t_{\text{logic cd.}}$$

Path-1) $\rightarrow \text{data} - CL_1 - CL_2 - CL_3 - CL_1$
 $\text{Reg} - CL_1 - \text{Reg} \rightarrow$

$$T_{\text{cycle}} = 50p + 500p + 50p = 600p \quad (12)$$

$$t_{\text{hold}} < 50 + 100 \quad (\text{No violation})$$

$\text{Reg} - CL_2 - \text{Reg} \rightarrow$

$$t_{\text{cycle}} = 50p + 800p + 50p = 900p \quad (12)$$

$$t_{\text{hold}} = 50p < 50 + 100p \rightarrow \text{No violation}$$

$\text{Reg} - CL_3 - \text{Reg} \rightarrow$

$$t_{\text{cycle}} = 50p + 600p + 50p = 700p \quad (12)$$

$$t_{\text{hold}} \Rightarrow 50p < 50 + 100 \quad (\text{No violation})$$

Path-2 $\rightarrow \text{data} - CL_1 - CL_2 \rightarrow CL_4 - CL_5 - \text{Reg}$

$\text{Reg} - CL_1 - \text{Reg}$

$$T_{\text{cycle}} = 600p \quad (12)$$

$$t_{\text{hold}} \Rightarrow 50p < 50 + 100 \quad (\text{No violation})$$

$\text{Reg} \rightarrow CL_2 \rightarrow \text{Reg} - CL_4 - \text{Reg}$

$$T_{\text{cycle}} = 50p + 800p + 500p + 50p = 1400p \quad (12)$$

$$t_{\text{hold}} = 50p < 50p + 100 + 100 \quad (\text{No violation})$$

$\text{Reg} - CL_5 - \text{Reg}$

$$T_{\text{cycle}} = 50p + 600p + 50p = 700p \quad (12)$$

$$t_{\text{hold}} \Rightarrow 50p < 50 + 100p \quad (\text{No violation})$$

Checking All Combinations $\rightarrow$

for correct operation $\rightarrow$

$$T_{\text{cycle min}} = 1400p$$

No thold violation at all.

$$\text{with skew } \pm 110 \text{ ps}$$

$$T_{\text{cycle}} \geq T_{c-q} + T_{\text{logic}} + t_{s.u}$$

$$t_{\text{hold}} < t_{c-q} + t_{\text{logic cd.}}$$

$$T_{\text{cycle}} = \dots = 490ps$$

$$t_{\text{hold}} = 50ps < 50 + 100 - 110p$$

$$T_{\text{cycle}} = \dots = 790ps$$

$$t_{\text{hold}} = 50p < 50p + 100p - 110p$$

$$T_{\text{cycle}} = 590ps$$

$$t_{\text{hold}} \Rightarrow 50p < 50 + 100p - 110p$$

$$T_{\text{cycle}} = 590ps$$

$$t_{\text{hold}} \Rightarrow 50p < 50 + 100p - 110p$$

$$T_{\text{cycle}} = 490ps, 710ps$$

$$t_{\text{hold}} = 50p < 50p + 100p - 110p$$

$$T_{\text{cycle}} = 1290ps, 1510p$$

$$t_{\text{hold}} \Rightarrow 50p < 50p + 100p + 100p$$

$$T_{\text{cycle}} = 590ps, 810ps$$

$$t_{\text{hold}} \Rightarrow 50p < 50p + 100p - 110p$$

$$t_{\text{hold}} \Rightarrow 50p < 50p + 100p - 110p$$

for correct operation.

$$T_{\text{cycle min}} = 1510p$$

t hold violation everywhere except at path Reg-CL2-c

Min. clock cycle time.

need to check for all paths (with skew/without skew)

$$T_{cycle_{min}} \geq T_{C-q} + T_{logic} + T_{S.U} \quad (2.5)$$

$$t_{hold} < t_{C-q} + t_{logic} + t_{cd}.$$

1-1) $\rightarrow data - CL_1 - CL_2 - CL_3 \rightarrow CL_1$
 $Reg - CL_1 - Reg. \rightarrow 2$

$$T_{cycle} = 50p + 500p + 50p = 600p$$

$$t_{hold} < 50 + 100 \quad (No violation) \quad (12)$$

- $CL_2 - Reg \rightarrow$

$$T_{cycle} = 50p + 800p + 50p = 900p \quad (12)$$

$$t_{hold} < 50p < 50 + 100p \rightarrow No violation$$

- $CL_3 - Reg \rightarrow$

$$T_{cycle} = 50p + 600p + 50p = 700p \quad (12)$$

$$t_{hold} \Rightarrow 50p < 50 + 100p \quad No (violation)$$

- 2 $\rightarrow data - CL_1 - CL_2 \rightarrow CL_4 - CL_5 - CL_1$

$\rightarrow CL_1 \rightarrow Reg$

$$T_{cycle} = 600ps$$

$$t_{hold} \Rightarrow 50p < 50 + 100 \quad No violation \quad (12)$$

$\rightarrow CL_2 \rightarrow Reg - CL_4 - Reg$

$$T_{cycle} = 50p + 800p + 500p + 50p = 1400ps \quad (12)$$

$$t_{hold} - 50ps < 50ps + 100 + 100 \quad No violation$$

- $CL_3 - Reg$

$$T_{cycle} = 50p + 600p + 50ps = 700ps$$

$$t_{hold} \Rightarrow 50p < 50 + 100ps \quad No violation \quad (12)$$

checking All Combinations $\rightarrow$

or correct operation $\rightarrow$

$$T_{cycle_{min}} = 1400ps$$

No thold violation at all.

with skew $\pm 110ps = \pm \delta$

$$T_{cycle} \geq T_{C-q} + T_{logic} + t_{su} \pm \delta \quad (2.5)$$

$$t_{hold} < t_{C-q} + t_{logic} + t_{cd} \pm \delta \quad (2.6)$$

$$T_{cycle} = 490ps, 710ps$$

$$t_{hold} = 50ps < 50 + 100 - 110p \quad Yes$$

(12)

+ve skew Neg. skew

$$T_{cycle} = 790ps, 910ps$$

$$t_{hold} 50p < 50p + 100p - 110p \quad Yes violation$$

-ve skew - No violation (12)

$$T_{cycle} = 590ps, 810ps$$

$$t_{hold} \Rightarrow 50p < 50 + 100p - 110p \quad Yes$$

$$-ve skew \rightarrow No violation \quad (12)$$

$$T_{cycle} = 490ps, 710ps$$

$$t_{hold} 50p < 50p + 100p - 110p \quad Yes violation$$

-ve skew $\rightarrow$ No violation.

$$T_{cycle} = 1290ps, 1510ps \quad (12)$$

$$t_{hold} \Rightarrow 50p < 50p + 100ps + 100ps - 110p \quad No violation$$

$$-ve skew \rightarrow No violation, \quad (12)$$

$$T_{cycle} = 590ps, 810ps$$

$$t_{hold} \Rightarrow 50p < 50p + 100ps - 110ps$$

-ve skew $\rightarrow$ No violation, Yes violation (12)

for correct operation.

$$T_{cycle_{min}} = 1510ps.$$

t hold violation everywhere except at path $Reg - CL_2 - CL_2 - Reg$.

(2)

PS $\Rightarrow$

a) $\tau_{PHL} = ?$

From I-V graph

$I_1 = 100 \mu A$, when $V_{DS} = 1V$, $V_{GS} = 1V$

Given $I_2 = 80 \mu A$, but propagation delay is referenced at 50% of V_{DD} . ($0.5 V_{DD}$)

$\therefore$ Current at 0.5V is

$$I_2' = 80 \mu A + \frac{(0.5 - 0.2) 20 \mu A}{(1 - 0.2)}$$

$$I_2' = 87.5 \mu A$$

$$I_{eq} = \frac{I_1 + I_2'}{2} = \frac{93.75 \mu A}{2} \quad (2)$$

$$\tau_{PHL} = \frac{C_L \Delta V}{I_{eq}}$$

$$= \frac{10 \text{ fF} \times (1 - 0.5)}{93.75 \mu A} \quad (1.5)$$

$$\tau_{PHL} = 53.33 \text{ ps}$$

$$\begin{aligned} \tau_{PLH} &= \frac{C_L \Delta V}{I_{eq}} \quad (1.5) \\ &= \frac{10 \text{ fF} \times (0.5)}{93.75 \mu A} \\ \tau_{PLH} &= 53.33 \text{ ps} \end{aligned}$$

c) R when $V_{DS} = 1V$

$$R_1 = \frac{1V}{100 \mu A} = 10 \text{ k}\Omega \quad (1)$$

from part ① $I_2' = 87.5 \mu A$

$$R_2 = \frac{0.5V}{87.5 \mu A} = 5.71 \text{ k}\Omega \quad (1)$$

$$\tau_{PHL} = \ln 2 \cdot R_{eq} \cdot C_L \quad (1)$$

$$R_{eq} = \frac{10 \text{ k}\Omega + 5.71 \text{ k}\Omega}{2}$$

$$R_{eq} = 7.86 \text{ k}\Omega \quad (1)$$

$$\tau_{PHL} = 54.23 \text{ ps}$$

$$\tau_{PLH} = 54.23 \text{ ps}$$

$$V_{th} = \frac{V_{to} + \sqrt{\frac{1}{k_R} (V_{DD} + V_{tp})}}{1 + \sqrt{\frac{1}{k_R}}}$$

$$= V_{to} + \sqrt{\frac{\mu_p \text{ Cox } \frac{W}{L}}{\mu_n \text{ Cox } \frac{W}{L}}} (1 + 0.4)$$

$$1 + \sqrt{\frac{\mu_p \text{ Cox } W_p}{\mu_n \text{ Cox } W_n}}$$

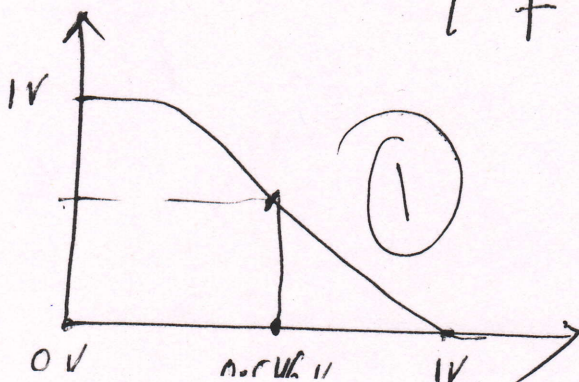
$$= 0.4 + \sqrt{\frac{40 \times 10^{-6} \times 3.94}{140 \times 10^{-6} \times 1.65}} (1.4)$$

$$1 + \sqrt{\frac{40 \times 10^{-6} \times 3.94}{140 \times 10^{-6} \times 1.65}}$$

$$V_{th} = \frac{0.4 + 0.171 \times 1.4}{1 + 0.171} = \frac{0.6394}{1.171}$$

$$1 + 0.171$$

$$1.171$$



$$V_{th} = 0.546$$

$$k_{eq} = \frac{31k\Omega}{W_p}$$

$$k_{eq} = \frac{13k\Omega}{W_n}$$

$$W_p = \frac{31k\Omega}{7.86k\Omega} = 3.94$$

$$W_n = \frac{13k\Omega}{7.86k\Omega} = 1.65$$

$$k_R = \frac{k_n}{k_p}$$

(2)

8.6

(i) We know that: $P \propto C V_{DD}^2 f$ and $f \propto \frac{1}{t_p} \propto \frac{I_{Dsat}}{C V_{DD}}$

$$\text{So, } P \propto I_{Dsat} V \propto k' \frac{W}{L} (V - V_t)^2 V \propto (s) \frac{(W)}{(\frac{L}{s})}$$

To keep power constant we need to scale $W \propto \frac{1}{s^2}$, which means redesigning gates with W a factor of $\frac{1}{s}$ smaller.

(ii)

$$t_p \propto \frac{C V}{k' \frac{W}{L} V^2} \propto \frac{W L \frac{C}{t}}{k' \frac{W}{L} V} \propto \frac{(\frac{1}{s^2}) (\frac{1}{s}) \frac{1}{1/s}}{s \cdot \frac{1/s^2}{1/s}}$$

$$\therefore t_p \propto \frac{1}{s^2}$$

(5m)

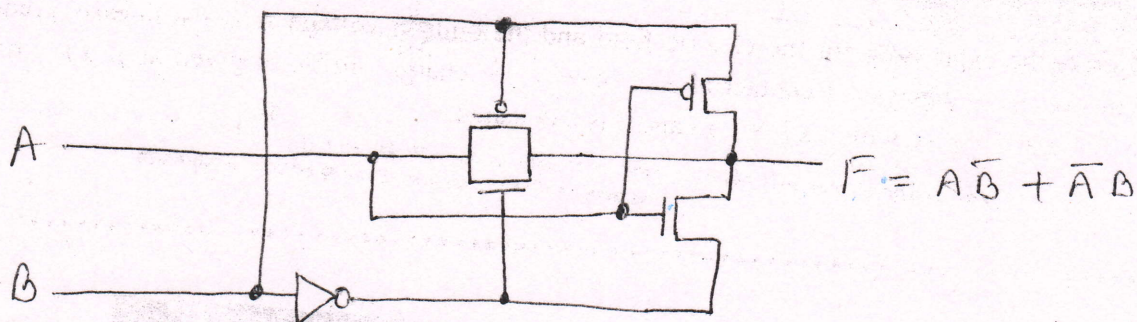
(iii) $P \propto I_{sat} V_{DD} \propto V_{DD} W C_{ox} (V_{gs} - V_t) V_{max} \propto W (s)$
So $W \propto \frac{1}{s}$

This means that no changes need to be made.

(5m)

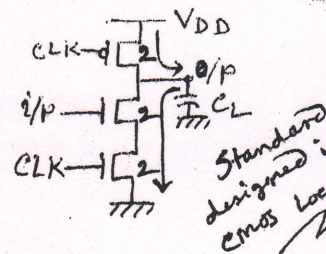
8.7

(i)

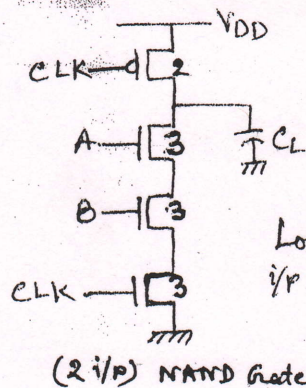
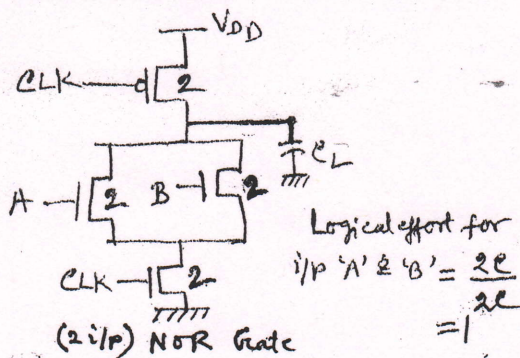


(5m)

Six transistor CMOS T2 implementation of the XOR function.



(11)
(b) "FALSE"



Logic for NMV high than the NOR (in dynamic logic)
NOR is over M

4m

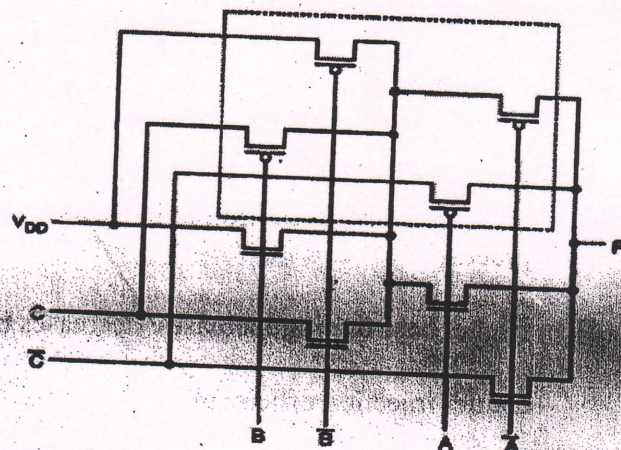
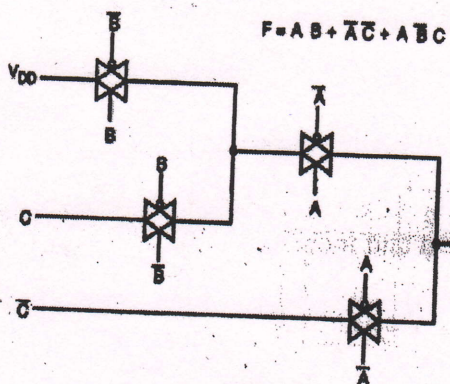
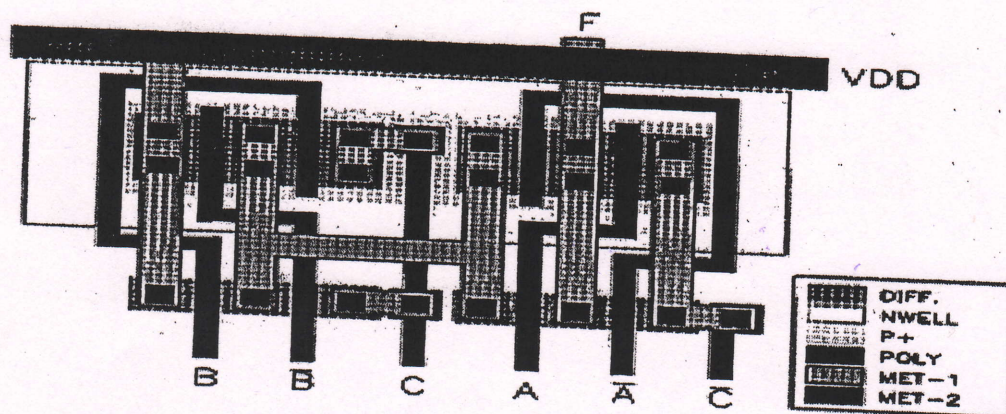


Figure 3. (a) CMOS TG realization of a three-variable Boolean function. (b) All pMOS transistors can be placed into one n-well to save area.



Mask layout of the CMOS TG circuit shown in Fig. (a)

6m